

Microstructure-efficient estimation of sovereign yield curves in less-liquid markets.

Glosten-Milgrom and Ho-Stoll foundations for liquidity-weighted NSS fitting, with asymptotics, identification and a 21-year Polish bond panel.

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Motivation and question

FROM PRACTITIONER PATCHES TO A DERIVED ESTIMATOR

- In a less-liquid sovereign market the fixing on a rarely traded bond is a far noisier signal than one on a benchmark issue: observation noise is strongly heteroskedastic across bonds.
- Practice has accumulated patches without foundations: turnover or outstanding weights tuned by hand, a τ_1 floor, a curvature ridge, a residual-maturity filter.
- The paper derives the efficient weight matrix for the Nelson-Siegel-Svensson fit from two dealer-market models, proves consistency, normality and Aitken efficiency, and turns the patches into corollaries of a rank analysis.
- Validated on a Monte Carlo grid and on 5,319 Polish daily cross-sections, 2005 to 2026.

The LW-NSS estimator

WEIGHTED NONLINEAR LEAST SQUARES IN YIELD SPACE

The zero curve follows Svensson (1994):

$$y(\tau; \theta) = \beta_0 + \beta_1 \frac{1 - e^{-\tau/\tau_1}}{\tau/\tau_1} + \beta_2 \left(\frac{1 - e^{-\tau/\tau_1}}{\tau/\tau_1} - e^{-\tau/\tau_1} \right) + \beta_3 \left(\frac{1 - e^{-\tau/\tau_2}}{\tau/\tau_2} - e^{-\tau/\tau_2} \right)$$

and the LW-NSS estimator solves, per settlement date,

$$\hat{\theta}_{\text{LW}} = \arg \min_{\theta \in \mathbf{■}} \sum_{i=1}^N w_{y,i} (y_i - \hat{y}_i(\theta))^2.$$

- Equal weights (Diebold-Li, Gürkaynak-Sack-Wright), duration weights and ad-hoc turnover blends coexist in practice. The open question is which $w_{y,i}$ a microstructure model actually implies.

Channel one: adverse selection (Glosten-Milgrom)

A YIELD-SPACE DEALER PROBLEM DELIVERS THE TURNOVER WEIGHT

- The fixing on bond i is the dealer's posterior mean of the fundamental yield after a Glosten-Milgrom quoting process run in the yield coordinate. In the active-trading regime the posterior variance is

$$\sigma_{y,i}^2 \approx \frac{\kappa}{\mu_i},$$

with μ_i the bond's turnover intensity. Duration does not enter the noise of a yield quote, it is only the price-to-yield Jacobian.

- **Theorem (optimal GM weight).** The efficient GLS weight is the inverse noise variance, $w_{y,i} = \mu_i/\kappa$: the turnover-weighted fit is minimum-variance unbiased within weighted NLS on yield residuals (Aitken).

Channel two and the hybrid: inventory risk (Ho-Stoll)

ONE DOMINANCE RATIO GOVERNS THE BLEND

- A complementary Ho-Stoll inventory channel delivers $w_{y,i,HS} = \text{out}_i / \kappa_{HS}$, with out_i the outstanding amount. With both channels active and additive,

$$\sigma_{y,i}^2 = \frac{\kappa_{GM}}{\mu_i} + \frac{\kappa_{HS}}{\text{out}_i} \quad \Rightarrow \quad w_{y,i,\text{hybrid}} = \frac{\mu_i \text{out}_i}{\kappa_{HS} \mu_i + \kappa_{GM} \text{out}_i}.$$

- The blend is governed by a single dominance ratio $\rho = \kappa_{HS} / \kappa_{GM}$, not separately identified at a single date but identifiable from a panel of dates by minimising leave-one-out yield-prediction error.
- The linear turnover-outstanding blend used heuristically in less-liquid markets is the first-order arithmetic-mean approximation to this harmonic-mean hybrid.

Asymptotic theory

M-ESTIMATION UNDER CONDITIONAL HETEROSKEDASTICITY

- The LW-NSS fit is a nonlinear M-estimator at fixed settlement date with $N \rightarrow \infty$, a thought experiment about a denser market that characterises the local behaviour of $\hat{\theta}$ on the observed panel.
- Consistency and asymptotic normality hold with sandwich variance

$$\sqrt{N}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, Q^{-1} \Omega Q^{-1}), \quad Q = \text{plim } \frac{1}{N} J' W J,$$

collapsing to Q^{-1} (up to scale) under correct specification.

- **Aitken efficiency.** The inverse-variance weight attains the Cramér-Rao style bound within the weighted NLS class, so any misspecified weight (equal weights included) is weakly less efficient, uniformly across GM, HS and hybrid. A robust variance estimator covers the misspecified case.

Identification under sparse panels

A FOUR-BAND COVERAGE CONDITION ON THE MATURITY GRID

- The Svensson loadings peak at $\tau \approx 1.7933 \tau_1$ and $1.7933 \tau_2$, which pins a four-band partition of the maturity axis:

$$B_1 = [0, \frac{\tau_1}{2}], \quad B_2 = [\frac{\tau_1}{2}, 2\tau_1], \quad B_3 = [2\tau_1, \frac{\tau_2}{2}], \quad B_4 = [\frac{\tau_2}{2}, \infty).$$

- **Proposition.** The information matrix $\frac{1}{N} J' W J$ is positive definite if and only if each band carries nonnegligible weight. Proof by Schur-complement partition into $(\beta_0, \beta_1, \beta_2, \tau_1)$ and (β_3, τ_2) blocks.
- **Corollaries.** When a band fails, a constrained fit fixing the corresponding decay at the boundary is consistent for the projection onto the restricted submanifold. The practitioner τ_1 floor and the curvature ridge are exactly these constrained estimators.

Monte Carlo design

CONTROLLABLE PANELS AROUND THE RANK-CONDITION BOUNDARY

- Mid-cycle Svensson truth $\theta_0 = (0.04, -0.01, 0.02, 0.01, 1.5, 5.0)$, pricing errors calibrated so a typical bond has $\sigma_{y,i} \approx 8$ bp, matching the Polish panel scale. 250 replications per cell, multi-start Levenberg-Marquardt.
- Grid crosses panel size $N \in \{12, 18, 24, 36\}$, three sparsity regimes (dense, moderate gap, severe gap with two of the four bands undercovered, mimicking the 2008-2009 Polish panel), and three liquidity-heterogeneity regimes (homogeneous, Pareto 80/20, extreme Pareto 95/10).
- Three noise DGPs (pure GM, pure HS, hybrid) against four estimators: E1 equal weights, E2 turnover (GM), E3 outstanding (HS), E4 hybrid with grid-searched dominance ratio.

Monte Carlo results

MEDIAN INTEGRATED YIELD-CURVE MSE, BASIS POINTS SQUARED

DGP	E1 (equal)	E2 (GM)	E3 (HS)	E4 (hybrid)
Pure GM noise	108.6	69.2	97.6	77.1
Pure HS noise	117.6	115.2	71.8	79.8
Hybrid GM+HS noise	131.0	100.6	98.0	91.2

- Equal weighting is uniformly worst, 50 to 60 percent above the channel-matched alternative. The channel-matched estimator wins under its own DGP, and the hybrid trails the winner by under 10 bp² everywhere while dominating under hybrid noise.
- The condition number of $J'WJ$ rises by a factor of about 50 from dense to severe-gap panels, validating the identification result.

| The Polish bond panel, 2005-2026

BONDSPOT QUOTE-DRIVEN PRIMARY-DEALER VENUE

- End-of-day fixings on 5,322 trading dates from 2005-01-03 to 2026-04-21. On a typical date 17.4 bonds are listed, of which on average 15.1 enter the fit. Three thin dates are dropped, leaving 5,319 daily cross-sections.
- Each bond carries a closing mid-price, a coupon-and-redemption schedule, a monthly turnover proxy and an outstanding face value, both disclosed by the Ministry of Finance.
- The minimum-residual-maturity filter (0.85 years before 2017-05-12, 1.20 years thereafter) excludes bonds distorted by the Treasury switch-auction option.
- Headline tests run on 60 dates sampled uniformly across the panel. The term-premium exercise uses the full panel.

Headline result: leave-one-out yield prediction

LOO MAE IN BASIS POINTS, 60 SAMPLE DATES

Statistic	E1 (equal)	E2 (GM)	E3 (HS)	E4 (hybrid)
Median LOO MAE (bp)	9.34	8.60	9.42	8.37
Mean LOO MAE (bp)	9.23	8.82	9.14	8.72
75th-percentile LOO MAE (bp)	12.61	11.99	12.60	11.99
Win rate vs E1 (%)	ref.	75.0	50.0	78.3

- The empirical pattern reproduces the simulation: equal weighting is uniformly worst, the hybrid weight cuts the median error by about one basis point and beats equal weights on 78 percent of dates.
- Wilcoxon signed-rank test on per-date differences, one-sided: $W = 439$, $p = 0.0002$ for E2 vs E1 on $n = 60$ dates, with an equally significant rejection for the hybrid.

When does the inventory channel matter?

REGIME-CONDITIONAL SPLIT BY THE GRID-BEST DOMINANCE RATIO

	GM-dominant ($n = 40$)	HS-leaning ($n = 20$)
Median 2y zero yield (%)	4.70	1.99
Median 10y zero yield (%)	5.57	3.41
Median 10y–2y slope (bp)	83	120
Calendar concentration	2005-2011, 2017-2026	2012-2016

- On GM-dominant dates pure turnover weighting matches the hybrid (under 0.1 bp apart). On HS-leaning dates the hybrid beats pure GM by about 0.7 bp median LOO MAE, the entire empirical edge.
- The inventory channel activates when yields are low and the curve is steep (2012-2016), adverse selection dominates otherwise. The split is robust to the ρ grid and the regime boundary.

Downstream: ACM term premia inherit the wedge

SAME FITTED HISTORIES, SAME ACM SPECIFICATION, ONLY THE WEIGHTS DIFFER

- Every cross-section is refitted under the four schemes and passed through the same Adrian-Crump-Moench model ($K = 5$ principal components, monthly resampling, daily projection).
- The 10-year term premium moves by 2.6 bp in median absolute difference between equal and hybrid weighting, 95th percentile 8.5 bp, mean signed difference zero: the weighting changes the day-to-day path, not the level.
- The wedge is widest when the panel was thinnest (2005-2011, 95th percentile 13.3 bp) and narrowest after 2017. Pure GM and hybrid are nearly indistinguishable (0.6 bp median).
- The wedge is the same order as the curve-fitting gain itself, as an order-one pass-through from the long fitted yield into the premium predicts.

Takeaways

PATCHES BECOME COROLLARIES

- Turnover weighting is the information-matrix optimum under adverse selection, outstanding weighting under inventory risk, their harmonic-mean hybrid under both.
- The estimator is consistent, asymptotically normal and Aitken efficient. Identification of the Svensson curvature is a four-band coverage condition that turns the τ_1 floor and the curvature ridge from folklore into constrained estimators.
- Equal weighting is uniformly the least accurate scheme in simulation and in 21 years of Polish data. The hybrid weight is the recommended default.
- Caveats: fixed-date asymptotics, linear adverse-selection scaling, dominance ratio identified by grid search rather than a structural moment.

Companion data and replication

YIELDCARTOGRAPHY.COM

- Every curve on the site is an LW-NSS fit, daily since 2005, with weights built from BondSpot turnover and outstanding amounts:
yieldcartography.com/curves.
- The ACM term premia built on these curves update daily on [/term-premia/](#), and per-bond rich/cheap diagnostics against the fitted curve are published on the experimental bond-level dashboards.
- The article is open access (CC-BY), and the Monte Carlo audit trail, with the frozen engine, configuration manifest and per-cell results, is captured as Supplementary Material to the paper.

Thank you.

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